

Mission 3 — Mechanical Design Project Build

Step 1: Plan Submission

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1. Project Overview

Project Name: Adjustable Phone/Tablet Stand

I will design a complete end-to-end mechanical assembly consisting of three separate parts — a base plate, a vertical support arm, and an adjustable bracket — joined together using assembly joints in Autodesk Fusion. This project simulates a real product design workflow and is suitable as a portfolio-grade deliverable.

2. Parts Breakdown

Part	Description	Dimensions
Base Plate	Flat weighted bottom with mounting holes	120mm x 80mm x 10mm
Vertical Arm	Upright support column	20mm x 20mm x 150mm
Bracket	Adjustable phone/tablet holder	80mm x 60mm x 5mm
Joints	Connect all parts together	M5 standard

3. Build Plan — Step-by-Step

Step 1: Design Base Plate: sketch 120mm x 80mm rectangle, add 4 corner holes, extrude to 10mm, fillet edges

Step 2: Design Vertical Arm: sketch 20mm x 20mm square, extrude to 150mm height, add bottom screw hole

Step 3: Design Bracket: sketch L-shaped profile 80mm x 60mm, extrude to 5mm, add adjustment slots

Step 4: Assemble: insert all 3 parts into one assembly file using Insert Component

Step 5: Apply Joints: connect arm to base (rigid joint), connect bracket to arm (slider joint for adjustability)

Step 6: Polish: apply fillets, set realistic material appearance, verify no gaps or overlaps

Step 7: Render: use Fusion Render workspace with Soft Box lighting for a professional image

Step 8: Document: write README, capture screenshots of each part and full assembly

Step 9: Submit: upload to GitHub repository with screenshots and description

4. Tools I Will Use

Tool	Purpose
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Autodesk Fusion	CAD modeling, assembly, joints, and rendering
GitHub	Documentation and final submission
Google Drive	Backup of screenshots and render images

5. Time Estimate

Phase	Estimated Time
Designing 3 Individual Parts	2 hours
Assembly & Joints	1 hour
Polish & Render	45 minutes
Documentation & Screenshots	45 minutes
GitHub Upload & Submission	30 minutes
Total	~5 hours